

Promptable Foundation Policies with Hilbert Representations: a general route to multi-objective RL

Problem and Motivation

Many real-world control tasks are inherently multi-objective: we must push primary performance while respecting safety, cost, or latency constraints. Retraining a new policy for every reward mixture is expensive and brittle. We propose a learn-once, prompt-later alternative. We first learn a representation ϕ that embeds states into a Hilbert space $\mathcal{Z}$ such that Euclidean distances approximate optimal temporal distances in the MDP. Then we train a single latent-conditioned policy $\pi(a | s, z)$ whose immediate reward is the inner product $\langle \phi(s') - \phi(s), z \rangle$. The vector z acts as a prompt: changing z changes the long-horizon direction of motion in $\mathcal{Z}$. At test time, new objectives are realized by solving for a new z , without re-optimizing π .

How prompting subsumes many objectives?

When ϕ preserves the geometry of optimal time-to-go, directions in $\mathcal{Z}$ correspond to meaningful long-horizon behaviors. The feature $\psi(s, a, s') = \phi(s') - \phi(s)$ summarizes a transition's direction; any downstream reward or constraint that can be expressed as a linear functional over such directions admits a latent prompt. Given samples (ψ_i, r_i) , we estimate z^* by least squares method, $z^* = \arg \min_z \sum (\langle z, \psi_i \rangle - r_i)^2$ and execute $\pi(\cdot | s, z^*)$. For goal-reaching we may set $z \propto \phi(g) - \phi(s)$; for safety, we choose z normal to the safe set embedding. In both cases the agent moves monotonically toward the desired set without retraining.

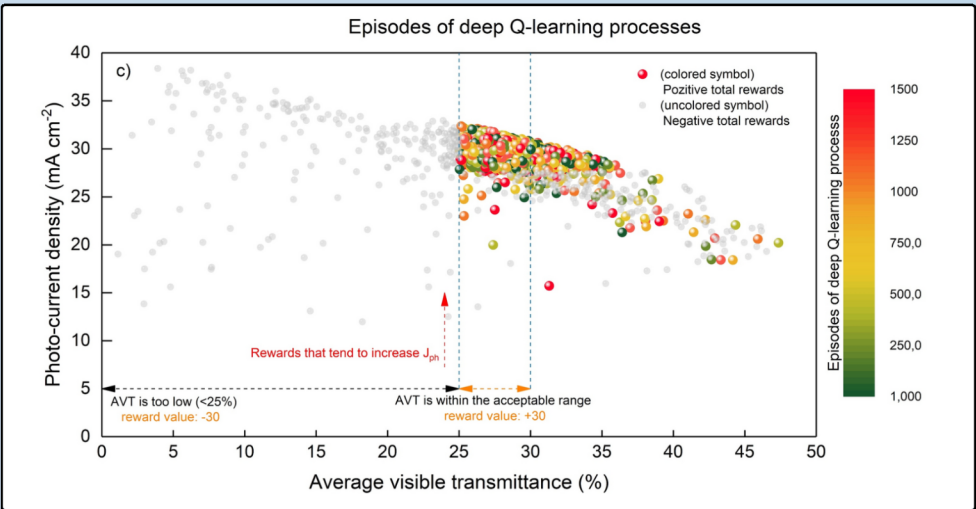
Training recipe

We learn ϕ using contrastive/time-ordering losses that favor distance preservation for near-future state pairs and separation for far-future ones. We then train π with random z directions so that it spans the cone of feasible motions in $\mathcal{Z}$. This pretraining is task-agnostic; task-aware adaptation reduces to solving a small regression for $\mathcal{Z}$, which is sample-light and stable.

- **Inputs.** Offline logs $D = (s, a, s')$.
- **Learn the geometry.** Train an embedding ϕ so that $\|\phi(s) - \phi(g)\|_2$ time-to-go; cache $\psi(s, a, s') = \phi(s') - \phi(s)$
- **Train one policy.** Optimize a single latent-conditioned policy $\pi(a | s, z)$ with reward $\langle \psi, z \rangle$ while sampling random z directions (so the policy can move along any direction in $\mathcal{Z}$).
- **Fit the objective vector.** For a new objective, solve once for $z^* = \arg \min_z \sum (\langle z, \psi_i \rangle - r_i)^2$ (ridge if needed).
- **Execute.** Run $\pi(\cdot | s, z^*)$

Assumptions and limitations

Performance hinges on ϕ 's ability to approximate optimal temporal distances; highly discontinuous dynamics may require localized embeddings or cone-restricted prompts. Coverage matters: if training saw too few directions, test-time prompts may extrapolate poorly. Hard constraints may need projecting $\mathcal{Z}$ into safe cones or augmenting the reward with barrier terms; both remain prompt-level operations, not policy retraining.



The figure placed to set a en example. it illustrates a structural tension between visible transmittance (AVT) and photo-current density. J_{ph} , which makes the task inherently MORL: there is no single “correct” reward, because stakeholders may prioritize different AVT targets (for instance, preference around 35%). Our approach treats preference as a vector prompt. We first learn an embedding. $\phi(s) \in \mathcal{Z}$ that approximates time-to-go geometry and train one latent-conditioned controller $\pi(a | s, z)$ with instantaneous reward $\langle \phi(s') - \phi(s), z \rangle$. A new preference is realized by solving once for a vector z (or setting it geometrically), then executing the same policy; scanning $\mathcal{Z}$ systematically explores the attainable trade-offs, revealing different Pareto-consistent behaviors—same policy, different prompts; the objective moves, not the policy.

This poster draws inspiration from Park, Kreiman, and Levine’s “Foundation Policies with Hilbert Representations” (ICML 2024)

S. Park, T. Kreiman, and S. Levine, “Foundation Policies with Hilbert Representations,” in Proc. 41st International Conference on Machine Learning (ICML), PMLR, vol. 235, pp. 39737–39761, 2024.